The Omega Architecture: Federated Genomics v4.2

Technical Whitepaper for Institutional Hub Partners

1. EXECUTIVE SUMMARY

FederiGene Omega represents the next evolution in distributed genomic analysis. By leveraging Federated Learning (FL) and Trusted Execution Environments (TEEs), the platform enables global collaboration on sensitive genomic datasets without compromising patient privacy or data sovereignty.

2. THE FEDERATED GENOMIC LEARNING (FGL) FRAMEWORK

Traditional research requires centralizing massive datasets, creating high security risks.
FederiGene Omega reverses this: Models travel to the data, not the other way around.

- Secure Aggregation: Only model weights (gradients) are uploaded, never raw VCF files.
- Node Sovereignty: Institutions maintain 100% control over local compute nodes.

3. INTEGRATED SECURITY STACK (THE OMEGA ENGINE)

3.1. TRUSTED EXECUTION ENVIRONMENTS (TEEs)

FederiGene utilizes Intel SGX and NVIDIA H100 to create hardware-encrypted enclaves.

3.2. DIFFERENTIAL PRIVACY (DP)

Noise is mathematically injected into updates to prevent genomic reverse-engineering.

3.3. BLOCKCHAIN LEDGER

Every interaction is recorded on an immutable audit trail for HIPAA/GDPR compliance.

4. INSTITUTIONAL HUB BENEFITS

- White-labeling support for specific research consortiums.
- Dedicated solution architects and 24/7 priority support.
- Multi-region synchronization for low-latency global updates.